and (4) personal protective equipment (e.g., respiratory protection, gloves, eye protection, hearing protection).

Crystalline Silica Exposure Limits

The NIOSH REL for respirable crystalline silica is 0.05 mg/m^3 as a time-weighted average (TWA) determined during a full-shift personal breathing zone sample. This REL is applicable for most workers who work up to a 10-hr workday during a 40-hr workweek to reduce the risk of developing silicosis, lung cancer, and other adverse health effects [NIOSH 2002].

The current OSHA PEL for respirable dust containing crystalline silica for the construction industry is measured by impinger sampling. In the construction industry, the PELs for cristobalite and quartz are the same.

Since the PELs were adopted, the impinger sampling method has been rendered obsolete by respirable dust gravimetric sampling [OSHA 1996]. OSHA currently instructs its compliance officers to apply a conversion factor when converting between gravimetric sampling and the particle count standard when characterizing construction operation exposures [OSHA 2008].

On September 12, 2013, OSHA published a Notice of Proposed Rulemaking (NPRM) for occupational exposure to respirable crystalline silica. The NPRM was published in the *Federal Register* and proposes a PEL of 0.05 mg/m^3 for respirable crystalline silica as an 8-hr TWA exposure [78 Fed. Reg. 56274 (2013)].

The ACGIH TLV for quartz and cristobalite (respirable fraction) is 0.025 mg/m^3 as an 8-hr TWA exposure [ACGIH 2013a]. The documentation to the TLV states that "it is the concern about fibrosis (silicosis) and the precedent inflammatory process resulting from silica exposures, and the association of inflammation and fibrosis with lung cancer that leads to this recommendation [ACGIH 2010]."

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Appendix A: Laboratory tracer gas test procedures to evaluate ventilation controls on asphalt pavement milling machines

These tracer gas laboratory test procedures were adopted from the document titled *Engineering Control Guidelines for Hot Mix Asphalt Pavers* [NIOSH 1997] and modified for asphalt pavement milling machines. These test procedures are the result of a collaborative effort by labor, industry, and government to reduce respirable crystalline silica exposure during asphalt pavement milling in highway construction.

Purpose

This test procedure is a first step in evaluating the capture efficiency of milling machine ventilation controls and is conducted in an indoor stationary environment before field testing. The goal of the test is to optimize the ventilation controls using tracer gas as a surrogate for respirable crystalline silica. This test can only approximate actual milling conditions, and actual milling conditions may give different results. The test will not evaluate filtration efficiency of ventilation controls if so equipped with a dust collection system.

Tracer gas studies have been completed for four milling machine manufacturers who are Partnership members. NIOSH technical reports from those studies can also be used to assist manufacturers of new asphalt milling equipment models in conducting tracer gas testing of their asphalt pavement milling machines [NIOSH 2011a, NIOSH 2011b; NIOSH 2012; NIOSH 2013].

Equipment/Facilities/Materials

- Two properly calibrated tracer gas monitors
- Asphalt pavement milling machine fitted with ventilation controls designed to prevent exposure to respirable crystalline silica
- Building with large opening (overhead door) to the outdoors
- Materials to construct a barrier, such as cardboard, plastic sheeting or tarpaulins, lumber of varying length, and duct tape
- Smoke generator
- PVC pipe: 5-centimeter (2-inch), schedule 40, 3 meters (10 feet) long, capped on one end, holes 6.4 millimeters (1/4 inch) in diameter drilled in a line every 15 centimeters (6 inches) on center (smoke distribution pipe)
- Tracer gas cylinder with dual stage pressure regulator fitted with an on/off valve after the second stage
- Zero air or nitrogen cylinder
- A tracer gas detector capable of measuring the selected tracer gas
- Polytetrafluoroethylene (PTFE) tubing: 3.2-millimeter (1/8-inch) outside diameter, 4 meters (13 feet) long
- Copper pipe: 12.7-millimeter (1/2-inch) inside diameter, the same length as the drum width, holes 0.79 millimeters (1/32 inch) in diameter drilled in a straight line 30 centimeters (12 inches) on center (tracer gas distribution pipe)

- Polyethylene (PE) tubing: 6.4-millimeter (1/4-inch) outside diameter, 30 meters (100 feet) long
- Two mass flow controllers with shut-off valves and a range appropriate for the selected tracer gas and zero air
- Bubble meter: 0.5-liter graduated cylinder, soap solution, and stop watch to measure gas flow rates; or a calibrated electronic flow calibrator
- Stainless steel tubing: 6.4-millimeter (1/4-inch) outside diameter, 30 centimeters (12 inches) long (sampling probe), sealed at one end, holes 1.6 millimeters (1/16 inch) in diameter drilled in a straight line 2.54 centimeters (1 inch) on center starting 2.54 centimeters (1 inch) from the sealed end (The number of holes will depend upon the diameter of the ventilation exhaust duct: a 20-centimeter [8-inch] exhaust duct requires the use of a sampling probe with seven holes 1.6 millimeters [1/16 inch] in diameter.)
- Manometer and pitot tube

Safety

The following safety precautions must be taken for each test:

- ✓ Use caution when handling smoke-generating equipment: it can be HOT!
- ✓ Make sure that the smoke generator does not set off fire sprinklers or create a false alarm.
- ✓ Avoid direct inhalation of smoke from the smoke generator. The smoke may act as an irritant in the upper respiratory tract.
- ✓ Transport, handle, and store all compressed gas cylinders according to the safety recommendations of the Compressed Gas Association (CGA) [CGA 2008].
- ✓ During testing, locate the compressed gas cylinder outdoors or in a well-ventilated area. The temperature should not exceed 125°F in the storage location.
- ✓ Avoid breathing the gas if a gas cylinder regulator malfunctions or some other major accidental release occurs. A major release of gas could displace the ambient air and reduce the oxygen level in the environment. Clear the area, and let the tank pressure reach equilibrium with the ambient environment. Wait until the concentration is below any applicable OEL before re-entering the area. Refer to the material safety data sheet (MSDS) or applicable OSHA PELs, NIOSH RELs or ACGIH TVLs for the safe level of the gas.

Procedures

General

These procedures are for testing the ventilation controls adapted to half-lane and larger milling machines for the control of respirable crystalline silica during asphalt pavement milling. These test procedures do not cover engine exhaust. The purpose of this testing is to optimize design parameters by evaluating different ventilation control flow rates.

Prepare the machine for testing as outlined below:

1. Position the milling machine in a bay area of a plant or garage with the outlet ducts placed through an opening leading to the outdoors.
2. Seal the building openings using cardboard, plastic, or other materials as shown in Figure A. This will help prevent tracer gas from recirculating back into the building.

3. Use wood, cardboard, or other materials to block off open areas between the cutting drum housing and the floor. These areas would normally be blocked by the ground during pavement milling due to the drum being below the surface of the road.
4. The indoor area surrounding the milling machine should be sufficiently large such that any tracer gas that escapes the drum housing is allowed to dissipate within the room without being drawn back into the drum housing area.
5. Position the cutting drum and conveyor as close as possible to the configuration used when milling. For example, the secondary conveyor should be at approximately the same height and angle as used in a typical pavement milling job.
6. Measure the duct velocity of the ventilation controls before and after tracer gas tests at each air flow rate. Information on how to measure duct velocity can be found in the "Testing and Measurement of Ventilation Systems" section in *Industrial Ventilation: A Manual of Recommended Practice for Operation and Maintenance* [ACGIH 2007].
7. During the performance evaluations, turn on the conveyors and set the engine to the speed typical for normal milling operations. The cutting drum should be rotating during the smoke and tracer gas tests to better replicate airflow patterns in the drum housing.



Figure A. Configuration for laboratory tracer gas test (Photo by NIOSH)

Smoke Test

The smoke test is designed to ensure that the barrier at the doorway is adequate for the quantitative tracer gas test and to visualize airflow patterns around the milling machine and conveyor areas. If smoke escapes from the drum housing or primary conveyor area, the air flow rate of the ventilation controls should be increased. If smoke leaks back into the building, seal any leaks in the barrier to the outdoors.

The sequence of a typical smoke test is outlined below:

1. Verify that the smoke generated by the test will not set off a fire alarm or fire-suppression system.
2. Place the 2-inch-diameter smoke distribution pipe directly beneath the cutter drum within 2 inches of the drum and secure in a horizontal position so that the holes are oriented upward on center with the cutter drum.
3. Use duct tape to block any holes in the smoke distribution pipe that extend beyond the width of the drum so that smoke is only released under the drum.
4. Connect the smoke generator to the smoke distribution pipe.
5. Clear the cutter drum area of any extraneous materials.
6. With the ventilation control on and the drum and conveyor running, release smoke in the drum area and look for leaks around the machine and through the barrier blocking smoke from coming in from the outdoors. Inspect the ventilation controls for unintended leaks at all fittings.
7. With the drum and conveyor running for a short time, deactivate the ventilation controls. This step is for simulating a no-control condition for comparison purposes.
8. Deactivate the smoke generator and wait for smoke levels to subside. If necessary to clear the room quickly, turn on additional exhaust ventilation.
9. Disassemble the test equipment (CAUTION!! The smoke generator may be HOT).
10. Repeat this test if substantial changes are made to the setup.

Tracer Gas Test

The tracer gas test is designed to calculate the volumetric air flow rate of the milling machine ventilation controls and to evaluate the system's effectiveness in capturing a surrogate contaminant under controlled indoor conditions. A tracer gas appropriate for ventilation testing should be used as the surrogate contaminant. A pitot tube traverse using procedures outlined in the *Industrial Ventilation Manual* should also be conducted to measure the ventilation system's air flow rate [ACGIH 2007]. The air flow rate determined from the pitot tube measurements should be compared with the tracer gas air flow rate calculations.

Ensure that the tracer gas detector has been calibrated according to manufacturer recommendations. Before testing begins, calibrate the mass-flow controllers with a bubble meter and a timer or with a calibrated electronic flow calibrator. Allow the tracer gas to flow through the pressure regulator to the mass-flow controllers to regulate the flow rate. Direct the flow through the wet bubble meter or calibrated electronic flow calibrator to determine the actual tracer gas flow rate [NIOSH 1973].

The 100% Capture Test

The purpose of conducting the 100% capture test is to provide another measure of the ventilation control air flow rate and to determine the concentration at which all of the tracer gas is captured for use in the capture efficiency calculation. The 100% capture test is completed by releasing a known rate of tracer gas into the ventilation control and sampling the diluted concentration of the tracer gas at a well-mixed location downstream in the ventilation exhaust duct. The accuracy of the mass flow controller calibration is pivotal to ensuring an accurate test. Releasing the tracer gas inside the intake to the ventilation controls will give the best chance for complete capture and mixing.

The sequence of the 100% capture test is outlined below:

1. Connect the supply tank of tracer gas to the release point using a pressure regulator, a mass-flow controller, and as much 6.4-millimeter (1/4-inch) tubing as needed to reach the tracer gas distribution pipe and the 100% tracer gas dosing (C100) locations.
2. Find a downstream sampling location in the duct where the tracer gas has been thoroughly mixed. If the sample is taken downstream of the fan, it should be sufficiently mixed. Long duct runs, sections of flexible duct, and duct fittings will also enhance mixing. If the gases are not thoroughly mixed, an extension on the exhaust duct may be needed.
3. If the air is thoroughly mixed, place a sampling hole 6.4 millimeters (1/4 inch) in diameter in the duct at least three duct diameters before the duct outlet. Insert the sampling probe through the sampling hole and position the holes in the sampling probe inside the duct and perpendicular to the direction of air flow. Secure the sample probe using tape. Sample tubing should connect the sampling probe to the detector and should be airtight to ensure that the sample is pulled from within the duct and not from the surrounding area.
4. Take at least five background measurements before starting the flow of tracer gas. Turn on the tracer gas and take at least five measurements during a 3-minute period and use the average of these concentration measurements to calculate the ventilation control air flow rate with Equation A-1. Before calculating the ventilation control flow rate ($Q_{(exh)}$), subtract background readings between sequential tests from the detector response.

$$Q_{(exh)} = \frac{Q_{(TG)}}{C_{(100)}} \times 10^6 \qquad Equation\ A-1$$

where

$Q_{(exh)}$ is the flow rate of air exhausted through the ventilation system (liters/minute or cubic feet/minute) (to convert from liters/minute to cubic feet/minute, divide liters/minute by 28.3),

$Q_{(TG)}$ is the flow rate of tracer gas (liters/minute or cubic feet/minute) introduced into the system, and

$C_{(100)}$ is the average concentration of tracer gas (parts per million) detected in the duct during the five 100% tracer gas capture tests.

5. If there is more than one ventilation duct, repeat the above procedure for each duct.

6. Following each tracer gas release, flush the tracer gas delivery tubing system with zero air (or nitrogen) for 1 minute.
7. Allow sufficient time between tests for the background readings to decay to less than 1% of the 100% tracer gas capture concentration.

Quantitative Capture Efficiency Test

The quantitative capture efficiency test is conducted to determine the capture efficiency of the ventilation control and uses the results of the 100% capture test.

The sequence of a quantitative capture efficiency test is outlined below:

1. Leave the sample probe, sample tubing, and tracer gas detector in the same location and orientation as the 100% capture test.
2. Leave the supply tank, pressure regulator setting, and mass flow controller setting in the same position as the 100% capture test.
3. The release location of the tracer gas supply tubing should be removed from the 100% location and connected to the 12.7-millimeter (½-inch) copper tracer gas distribution pipe.
4. Position the tracer gas distribution pipe under the cutter drum and orient the holes of the distribution pipe up toward the center axle of the cutter drum.
5. Start the milling machine engine and set the engine at the speed (rpm) typical for normal milling operations. Start the ventilation control and begin monitoring for tracer gas to determine background levels.
6. To determine background concentration, take at least five measurements during a 3-minute period.
7. Release the tracer gas at the same mass flow controller setting used in the 100% capture test. The tracer gas distribution pipe is designed to release the tracer gas evenly throughout the width of the cutter drum area.
8. Wait 30 seconds for the tracer gas concentration to reach steady state then take at least five measurements during a 3-minute period and calculate an average concentration. Subtract background concentration, and calculate the capture efficiency (η) using Equation A-2.

$$\eta = \frac{C_{(DRUM)}}{C_{(100)}} \times 100\% \qquad Equation\ A-2$$

where

η is the capture efficiency (%),

$C_{(DRUM)}$ is the average concentration of tracer gas (parts per million) detected in the duct when tracer gas is released in the drum housing, and

$C_{(100)}$ is the average concentration of tracer gas from the 100% capture test.

9. Discontinue the tracer gas flow and flush the tracer gas distribution system with zero air (or nitrogen) for 1 minute.

10. Allow sufficient time between tests for the background readings to decay to less than 1% of the 100% tracer gas capture concentration.
11. Perform a total of five pairs of the 100% capture tests and capture efficiency tests and determine the overall capture efficiency of the ventilation control from the average of all five trials.

Analysis of Tracer Gas Data

The tracer gas data should be analyzed to determine if the average capture efficiency is sufficiently large to conclude that the true mean capture efficiency exceeds 90% with 95% confidence. This approach treats the efficiency values as a normally distributed random sample. The lower confidence limit for the true efficiency is obtained by using the t distribution [Agresti and Franklin 2009].

To accomplish this, compute the lower 95% confidence limit for the mean capture efficiency and compare it to 90%; if the lower limit exceeds 90%, the statement can be made that the average efficiency exceeds 90% with 95% confidence.

The steps to carry out the analysis include:

1. Calculate the average (AVG) and standard deviation (STD) of the n efficiency values where n=number of independent observations
2. Calculate the degrees of freedom = n-1 and determine the 95th percentile of the t-distribution with n-1 degrees of freedom (i.e., t0.95, df=n-1)
3. Calculate the lower bound

   $$L = AVG - (STD/sqrt(n)) \times t(0.95, df=n-1) \qquad (A\text{-}3)$$

4. If the value L is greater than 90% then the system meets the 90% efficiency criterion.

In step 2, the 95th percentile can be obtained using a published table of t-distribution critical values or by using the "tinv(p,df)" function available in Microsoft EXCEL. The "tinv" function returns the two-tailed inverse of the t-distribution, so care must be exercised in specifying the parameters. The first parameter is the sum of the upper and lower tail probabilities (so for the 95% percentile, use p=0.05+0.05=0.1); the second parameter is the number of degrees of freedom (i.e., df=n-1).

Microsoft EXCEL can be used to analyze the tracer gas data, as illustrated with the following example data.

Example: Suppose that five tests are done and the five example data efficiencies are 93, 93, 94, 95, and 95. These are hypothetical values for the purpose of demonstrating the calculations. Using Equation A-3:

1. The average is 94 and the standard deviation is 1.
2. The degrees of freedom is 5-1=4 and the t-value is 2.132 (obtained in Excel by specifying tinv(0.1,4)).
3. The square root of 5 is 2.236 and the lower bound L=94-(1/2.236)×2.132=93.05.
4. Because L exceeds 90%, the statement can be made that the efficiency is greater than 90% at the 95% confidence level.

The sample calculation is shown using Microsoft EXCEL in Table A-1 to show equations and Table A-2 to show results.

Table A-1. Microsoft Excel example equations for determining L

	A	B
1	Trial	Capture efficiency
2	1	0.93
3	2	0.93
4	3	0.94
5	4	0.95
6	5	0.95
7	Number of trials (N)	5
8	average	=AVERAGE(B2:B6)
9	standard deviation	=STDEV(B2:B6)
10	confidence level	0.95
11	L	=B8-B9/SQRT(B7)*TINV(((1-B10)*2),(B7-1))

Table A-2. Microsoft Excel example results for determining L

	A	B
1	Trial	Capture efficiency
2	1	93%
3	2	93%
4	3	94%
5	4	95%
6	5	95%
7	Number of trials (N)	5
8	average	94.0%
9	standard deviation	0.01
10	confidence level	95%
11	L	93.05%

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Appendix B: Field Test Procedures

Field testing of the asphalt milling machines should be done after laboratory tracer gas tests of the ventilation controls show greater than 90% capture efficiency. The purpose of field testing is to provide information to the milling machine manufacturer about the performance of their engineering controls on typical highway construction sites. Asphalt milling machine manufacturers need reliable air sampling data for crystalline silica exposures to confirm the effectiveness of the engineering controls before those controls are placed on an entire fleet of milling machines. The procedure outlined in this appendix can also be used to verify the effectiveness of the overall respirable crystalline silica control strategy, which may include any combination of water sprays, ventilation, or other control methods.

Site Selection Considerations

Selected sites should be representative of typical asphalt pavement milling jobs in highway construction. To provide a rigorous test of the milling machine dust controls, manufacturers should consider the following parameters when selecting from a population of typical highway construction study sites for collecting air samples during asphalt pavement milling:

- Consider selecting sites where the milling crew will be working at least an 8-hour shift and less than 30% of the shift is spent on unexpected downtime (e.g., unusually long dump truck delays, milling machine repairs, and weather delays). Expected downtime such as refilling the milling machine with water, replacement of cutting teeth, switching between dump trucks when a truck is fully loaded with recycled asphalt pavement, and minor traffic delays are all typical to asphalt pavement milling and should not be considered as unexpected downtime.
- Do not select sites where more than half of the sample time includes milling around manhole covers and/or within busy intersections.
- Do not conduct air sampling in the rain or when the pavement is still wet from rain.
- Consider finding additional test sites if the average wind speed over any evaluated shift is greater than 5 m/s (11 mph).
- Consider finding additional test sites if the average crystalline silica content in the bulk material is less than 5%.
- Document other potential silica dust sources on or near the site, such as other milling machines without ventilation controls, soil stabilization, and farming operations. Sites should not be excluded based on the presence of other dust sources, but noting their presence may be useful when interpreting results.

Study Design and Data Collection

To field test the effectiveness of dust controls on asphalt pavement milling machines, personal breathing zone air samples of the pavement milling machine operator and a ground person should be analyzed for respirable crystalline silica. The air sample should be collected during a full shift of normal employee work activities. The workers' arithmetic mean exposures will be compared to the exposure limit. Crystalline silica exposures will be considered acceptably controlled if the upper 95% confidence limit for the arithmetic mean exposure associated with a randomly chosen day at a randomly chosen site is less than the exposure limit (e.g., 0.05 mg/m^3 for the NIOSH REL).

Because the exposure data are typically log-normally distributed, the estimated log scale means and standard deviations are needed to get upper confidence limits (UCL) for the arithmetic means (AM).

Sampling should be conducted on each of the two workers (operator and ground person) using one of the following two sampling strategies:

1. Sample three days at a minimum of three different milling sites (at least nine sampling days). If additional sites are added, three sampling days should be conducted at each additional site.
2. Sample one day at a minimum of five sites.

The statistical methods proposed here work best if a) the same individual does the job at each site or b) different workers do the job at different sites. This applies for both the operator and ground person.

Selection of the sampling strategy should be based on several considerations, including availability of suitable sampling sites, site-to-site variability, and within-site variability. The first strategy, sampling for three days at a minimum of three sites (at least nine sampling days), would be preferred if within-site variability is a major concern. The second sampling strategy, one sampling day at a minimum of five sites, would be preferred if the site-to-site variability is of primary concern. Availability of suitable sites, such as milling sites large enough to support three days of sampling, will also need to be factored into the sampling strategy decision.

Exposure concentrations should be compared against occupational exposure limits for respirable crystalline silica, such as the NIOSH REL of 0.05 mg/m^3, the ACGIH TLV of 0.025 mg/m^3, and the OSHA PEL. The exposure assessment should be performed by a safety and health professional familiar with the practice of industrial hygiene.

Air sampling should include time-integrated cyclone-filter-pump samples for assessment of workers' full-shift, time weighted average (TWA), personal breathing zone exposures to respirable crystalline silica. NIOSH has conducted this testing using BGI model GK2.69 cyclones (BGI Inc., Waltham, MA) at a flow rate of 4.2 liters/minute (L/min) with battery-operated sampling pumps. Sampling pumps should be calibrated before and after each day's use. The 4.2-L/min cyclone is suitable for this application because it should allow for a minimum detectible concentration of about 5% of the REL (10% of the TLV), assuming a 5-µg/sample laboratory limit of detection for quartz and an 8-hour sample time. Other cyclones operating at lower flow rates may not be suitable for this testing because of their higher minimum detectable concentrations.

A sampling pump should be clipped to each sampled employee's belt, worn at their waist. The pump should be connected via Tygon® tubing and a tapered Luer-type fitting to a pre-weighed, 37-mm diameter, 5-micron (µm) pore-size polyvinyl chloride filter supported by a backup pad in a three-piece filter cassette, sealed with a cellulose shrink band (according to NIOSH Methods 0600 and 7500) [NIOSH 1998, 2003]. The front portion of the cassette is removed and the cassette is attached to the BGI respirable dust cyclone.

Start and stop times for the sampling pump should be noted. This would include any time when the worker needed to remove the sampler, such as during lunch breaks. Significant events such as process upsets or sampling equipment failures should also be noted. However, pumps should not be stopped for normal delays that are common in asphalt pavement milling such as refilling the milling machine with water, replacement of cutting teeth, switching between dump trucks when a truck is fully loaded with recycled asphalt pavement, short breaks for workers to rehydrate, and minor traffic interruptions or delays.

Bulk samples of the asphalt pavement material should be collected. Collecting bulk samples is important to determine the percent silica in the asphalt pavement being milled, which will then be compared to the percent silica in the filter samples.

Filters should be analyzed gravimetrically for all respirable particulate using NIOSH Method 0600 and then by X-ray diffraction (XRD) for respirable crystalline silica using NIOSH Method 7500 [NIOSH 1998, 2003]. Bulk samples should be analyzed in accordance with NIOSH Method 7500. Analysis should be conducted by an AIHA-accredited laboratory.

Wind velocity, temperature, and humidity should be recorded continuously on-site during sampling. These data can be measured and recorded using any of a number of commercially available weather stations. It is important to locate the wind instrumentation in a location where the readings will not be influenced by passing traffic. It is also possible to use historical weather observations from the National Oceanic and Atmospheric Administration (NOAA) fixed weather station nearest to each evaluated site.

Data Analysis

The testing goal is to obtain an upper 95% confidence limit for the arithmetic mean exposure associated with a randomly chosen day at a randomly chosen site. If this upper 95% confidence limit is less than the occupational exposure limit then the exposures are said to be acceptably controlled. The exposure limit discussed here is the NIOSH REL, 0.05 mg/m^3.

Lognormality is typically assumed for exposure data. Lognormality means that the natural logarithms of the measurements approximately follow a normal distribution. The statistical literature provides a variety of methods for estimating an upper confidence limit for the arithmetic mean of a lognormal distribution. The method used here will be presented in two versions to correspond to the two proposed sampling strategies.

An Excel spreadsheet was developed so that manufacturers can analyze their own data to determine the level of dust control ("mover_upper_confidence_limits.xlsx").

Sampling Strategy 1

With sampling strategy 1, personal breathing zone air samples for respirable crystalline silica are collected for three days at each of three or more sites (ideally four or more).

The statistical method used for the analysis is the "MOVER" method which stands for "Method of Variance Estimates Recovery". The MOVER method can be used for the balanced design

(i.e., same number of days at each sampled site) but can also be used for the unbalanced design (i.e., different number of days at each sampled site). The MOVER method estimates the upper confidence limit (UCL_M) using an approach developed by [Zou et al. 2009].

Analysis of balanced data collected under sampling strategy 1 can be completed in the "Ex. Strategy 1-Balanced" worksheet of the attached Excel spreadsheet.

In the worksheet each measurement is converted to the natural log scale, and the mean for each job category (operator or ground person) is computed for each site. Use of the spreadsheet is described in Appendix D. The following quantities are computed in the spreadsheet:

$\bar{\bar{x}}$ = mean of the site means $(\bar{x}_s)$, for log scale data, which is the natural log of the geometric mean(GM)

$s_{\bar{x}_S}$ = standard deviation of the site means, for log scale data; it is the natural log of the geometric standard deviation for the lognormal distribution associated with the site means

n_{days} = number of days at each site

n_{sites} = number of sites

df1= n_{sites} -1, the degrees of freedom for the standard deviation of site means

df2= $n_{sites} \times (n_{days} - 1)$, for balanced data (same number of days at each site), the degrees of freedom for the within-site standard deviation

$astd_{ws}$ = average standard deviation within sites (square root of the average of the squares of the within-site standard deviations, when data are balanced).

Exponentiating this value gives the within-site (or between days) geometric standard deviation.

$\sqrt{s^2_{\bar{x}_s} - \frac{1}{n_{days}} astd^2_{ws}}$ =standard deviation between sites, which, in general, is smaller than the standard deviation between-site means, $s_{\bar{x}_S}$. Exponentiating this value gives the between-site geometric standard deviation.

To calculate the 95% upper confidence limit, first calculate UCL2, given by:

$$UCL2 = \bar{\bar{x}} + 0.5 s^2_{\bar{x}_s} + 0.5 \frac{n_{days} - 1}{n_{days}} astd^2_{ws} + \sqrt{\left(s_{\bar{x}_s} \times \frac{t(0.95,\ df1)}{\sqrt{n_{sites}}}\right)^2 + 0.5^2\ s^4_{\bar{x}_s} \left(\frac{df1}{chisq(0.05, df1)} - 1\right)^2 + \left(0.5 \times \frac{n_{days} - 1}{n_{days}}\right)^2 astd^4_{ws} \left(\frac{df2}{chisq(1 - 0.05, df2)} - 1\right)^2}$$

(B-1)

In equation (B-1), t(0.95,df1) is the 95th percentile of the t distribution with df1 degrees of freedom and chisq(1-0.05,df2) denotes the 95th percentile of the chi square distribution with df2 degrees of freedom.

Then the 95% upper confidence limit is given by

$UCL_M = e^{UCL2}$ (B-2)

This limit is discussed in more detail in Appendix C. Simulations in that section indicate that for most parameter values studied, the coverage for the method is close to the nominal 95%.

Analyses of Example Data

The following example data in Table B-1 are also in the "Ex. Strategy 1-Balanced" worksheet of the spreadsheet. For simplicity, the example considers three days of sampling at each of three sites rather than the recommended four sites. These are hypothetical values for the purpose of demonstrating the calculations.

Table B-1: Example data for 3 days per site at 3 sites

Site	Day	Groundperson silica concentration (mg/m³)	Ln (Ground person silica concentration)	Ground person site means, Ln scale		Within- site standard deviation, Ln scale
1	1	0.00507	-5.284	-5.645		0.354*
1	2	0.00349	-5.658			
1	3	0.00250	-5.992			
2	1	0.01200	-4.423	-5.172		0.692
2	2	0.00308	-5.787			
2	3	0.00495	-5.308			
3	1	0.01010	-4.595	-4.565		0.476
3	2	0.00575	-5.159			
3	3	0.01480	-4.213			
		$\bar{\bar{x}}$ =mean(site means)=		-5.158	$astd_{ws}$=average standard dev. (within-site)=	0.526**
		$s_{\bar{x}_S}$=standard dev.(site means)=		0.495***		
		df1=degrees of freedom=n_{sites}-1		2		

* 0.354 is the standard deviation of (-5.284,-5.658,-5.992).

** It is customary to average the variances, which are the squares of the standard deviations. The average within-site variance is 1/3 x (0.354^2 + 0.692^2 + 0.476^2)=0.277, the square root of which is 0.526. However, in the spreadsheet, the calculations use a more general form also applicable to unbalanced data: multiply each within-site variance by its degrees of freedom, n_{days}-1. Divide the sum of the weighted variances by the sum of the degrees of freedom: (2×0.354^2 + 2×0.692^2 + 2×0.476^2)/(2+2+2)=0.277.

*** 0.495 is the standard deviation of (-5.645,-5.172,-4.656).

The between-site standard deviation is given by $\sqrt{s^2_{\bar{x}_s} - \frac{1}{n_{days}} astd^2_{ws}}$ =sqrt(0.495^2-0.3333×0.526^2)=0.391.

Consequently, the within-site GSD is exp(0.526) = 1.692 and the between-site GSD is exp(0.391) = 1.48. Also, the GM is exp($\bar{\bar{x}}$)=exp(-5.158)=0.0058, where "exp" denotes the exponential function.

From equation (B-1), UCL2 = (Term1 + sqrt (Term2 + Term3 + Term4)).

Where:

Term1 = $\bar{\bar{x}} + 0.5 s^2_{\bar{x}_s} + 0.5 \frac{n_{days} - 1}{n_{days}} astd^2_{ws}$. Using values from Table B-1, Term1 = -5.158 + 0.5 x (0.495^2 + 2/3 x 0.526^2) = -4.9433. (The total ln scale variance is 0.495^2 + 2/3 x 0.526^2=0.429, which corresponds to a total ln scale standard deviation of $\sqrt{0.429}$=0.655. The total geometric standard deviation is $e^{\sqrt{0.429}}$=1.925. Also, note that the fraction of total variance due to sites is 0.391^2/0.429=0.36, where 0.391 is the between-site standard deviation.)

Term2 = $(s_{\bar{x}_s} \times \frac{t(0.95,\ df1)}{\sqrt{n_{sites}}})^2$ uses the upper confidence limit for the mean. Using values from Table B-1, Term2 = (0.495^2 x 2.920^2/3) = 0.6958, where df1 = n_{sites}-1 = 3-1 = 2 and 2.920 is the 95th percentile of the t distribution with 2 degrees of freedom.

Term3 = $0.5^2 s^4_{\bar{x}_s} (\frac{df\ 1}{chisq(0.05, df1)} - 1)^2$ uses the upper confidence limit for the between-site variance. Using values from Table B-1, Term3 = 0.5^2 x 0.495^4 x $(2/0.1026-1)^2$ = 5.125, where df1 = nsites-1 = 3-1 = 2 and 0.1026 is the 5th percentile of the chi square distribution with 2 degrees of freedom.

Term4 = $(0.5 \times \frac{n_{days}-1}{n_{davs}})^2 astd^4_{ws} (\frac{df\ 2}{chisq\,(1-0.05, df2)} - 1)^2$ uses the lower confidence limit for the average of the within-site variances (the reason for this is discussed in Appendix C). Using values from Table B-1, Term4 = 0.5^2 x 0.667^2 x 0.526^4 x $(6/12.592-1)^2$ = 0.0023, where df2 = n_{sites} (n_{days}-1) = 3(3-1) = 6, and 12.592 is the 95th percentile of the chi square distribution with 6 degrees of freedom.

Substituting these values into equations (B-1) and (B-2), gives UCL_M=exp(-4.9430 +sqrt(0.6958+5.125+0.0023))=0.0797 mg/m^3.

Thus, for the example data, the 95% upper confidence limit for the arithmetic mean for the ground person exceeds the NIOSH REL of 0.05 mg/m^3.

Unbalanced Data

When the number of days at each site is not the same, many of the terms in equation (B-1) remain the same; however, there are four important changes. Using the notation of [Krishnamoothy 2014]: $\bar{\bar{x}}$ is still the average of the site means, $s_{\bar{x}_s}$ is still the variance of the site means about $\bar{\bar{x}}$ and df1 is still (n_{sites}-1). The three changes are:

1) df2= (number of measurements)-(number of sites)

2) $astd_{ws}^2$ is the sum of the weighted within-site variances divided by df2. Each weight is the number of days at the site minus 1. For instance, if there are 3 days at sites one and two and 2 days at the third site, the weights are, respectively, 2 and 2 and 1.

3) In equation (B-1), $\frac{n_{days}-1}{n_{days}}$ is replaced by (1-nbar), where nbar is the average of the reciprocals of the number of days at the sites. For the example in (2) above, nbar = (1/3+1/3+1/2)/3 = 0.389. Thus, (1-nbar) = 1-0.389 = 0.611.

4) $\sqrt{s_{\bar{x}_s}^2 - \text{n bar x } astd_{ws}^2}$ =standard deviation between sites; note that for balanced data, nbar=1/n_{days}.

Analyses of Unbalanced Example Data

The data are shown in the worksheet "Ex. Strategy 1-Unbalanced". For the data used above, the measurement for site 2, day 3 has been omitted.

First, when the data are unbalanced, the squared within-site standard deviations must be weighted by their degrees of freedom. For the ground person data, the weights for the three within-site variances are 2, 1, and 2:

$astd_{ws}$ = sqrt[(2x0.12539 + 1x0.9292 + 2x0.2262)/(2+1+2)]

= sqrt[(0.2508 + 0.9292 + 0.4524)/5]=0.5714.

Second, the other important differences from the balanced case are due to the unequal number of days at each site. Let nbar=0.389, as was explained above. In equation (B-1), the term $\frac{n_{days}-1}{n_{days}}$ is replaced by 1-nbar=1-0.389=0.611. Also, df2=8-3=5, since there are eight measurements and 3 sites.

Also, $\sqrt{s_{\bar{x}_s}^2 - \text{nbar x } astd_{ws}^2}$ =standard deviation between sites=sqrt(0.4953^2-0.389×0.5714^2)=0.3440.

With these changes, the results are:

Term1 = $\bar{\bar{x}} + 0.5s_{\bar{x}_s}^2 + 0.5x(1 - \text{nbar})x astd_{ws}^2$ = -5.1349 + 0.5 x (0.4953^2 + 0.611 x 0.5714^2) = -4.9125. (The total ln-scale variance is 0.4953^2 + 0.611 x 0.5714^2=0.445, which corresponds to a total ln-scale standard deviation of √0.445=0.667. The total geometric standard deviation is $e^{\sqrt{0.445}}$=1.95.)

Term2 = $(s_{\bar{x}_s} x \frac{t(0.95, df1)}{\sqrt{n_{sites}}})^2$ = (0.4953 x 2.920/sqrt(3))^2 = 0.6972.

Term3 = $0.5^2 s_{\bar{x}_s}^4 \left(\frac{df\ 1}{chisq(0.05, df1)} - 1\right)^2$ $= 0.5^2 \times 0.4953^4 \times (2/0.1026 - 1)^2 = 5.146.$

Term4 = $(0.5 \times (1 - nbar))^2 astd_{ws}^4 \left(\frac{df\ 2}{chisq(1 - 0.05, df2)} - 1\right)^2$ =

0.5^2 x 0.611^2 x 0.5714^4 x $(5/11.0705 - 1)^2$ = 0.002992, where 11.0705 is the 95th percentile value of the chi square distribution with 5 degrees of freedom.

UCL_M = exp(-4.9125 + sqrt(0.6972 + 5.146 + 0.002992)) = 0.0825 mg/m³.

For both occupations the limits increase slightly when the third day at site 2 is omitted.

Conclusions from Data Analysis

As shown in the "strategy 1-balanced" worksheet, the 95% upper confidence limit for the operator is 0.177 mg/m³. Thus, for analysis of the full example data set at the 95% confidence level, it cannot be concluded that the exposures are controlled below the NIOSH REL of 0.05 mg/m³. This conclusion applies to the population of sites from which the sample sites were drawn, with 95% confidence for each job category, or 90% overall confidence for both job categories (operator and ground person).

If the upper confidence limits are above the NIOSH REL, this means one of two things, either (a) the population mean is above the REL and the dust control system is not capable of reducing exposures below the REL, or (b), the population mean is below the REL, but the sample size (3 days of sampling at each of 3 sites) does not permit a precise estimate of the population mean.

There is some additional discussion in the "Design Considerations" section after the discussion of sampling strategy 2.

Sampling Strategy 2

An alternative collection strategy involves sampling at five or more sites with one day per site.

Analysis of the data is done in the Excel spreadsheet with the worksheet name "Ex. Strategy 2". For sampling strategy 2, a minimum of five sites should be sampled, with only one day of sampling at each of the five sites.

$$UCL2 = \bar{x} + 0.5 s_x^2 + \sqrt{\left(s_x \cdot \frac{t(0.95, df1)}{\sqrt{n_{sites}}}\right)^2 + 0.5^2 s_x^4 \left(\frac{df\ 1}{chisq(0.05, df1)} - 1\right)^2} \qquad \text{(B-3)}$$

where df1=n_{sites}-1, $\bar{x}$ is the natural log scale mean, s_x^2 is the natural log scale variance, df1 is n_{sites}-1, $t(0.95, n_{sites} - 1)$ is the 95th percentile of the t distribution with df1 degrees of freedom, and chisq(0.05,df1) is the 5th percentile of the chi square distribution with df1 degrees of freedom.

$UCL_M = e^{UCL2}$ (B-4)

The following example data in Table B-2 are in the "Ex. Strategy 2" worksheet of the spreadsheet. These are hypothetical values for the purpose of demonstrating the calculations.